

Methods and Philosophy of Statistical Process Control

The basis of the control chart, control limits, ARL & ATS, and rational subgroups

Slides covered: **3-10, 21-25, 35-45**

Summary + all formulas & examples

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1 The Magnificent Seven

slides 3-10

To meet or exceed customer expectations, a product should be produced by a process that is **stable / repeatable**. SPC can be applied to any process; its seven major tools — “**the magnificent seven**” — are:

#	Tool	Purpose
1	Histogram / stem-and-leaf plot	Show the shape and spread of the data.
2	Check sheet	Organized collection of defect data over time.
3	Pareto chart	Rank defects in descending order to find the "vital few."
4	Cause-and-effect diagram	Find the root causes of a problem (fishbone).
5	Defect concentration diagram	Mark where defects occur on the product.
6	Scatter diagram	Study the relationship between two variables.
7	Control chart	Monitor a process over time and detect out-of-control conditions.

Note

Of the seven, the **Shewhart control chart** is probably the most technically sophisticated.

Constructing a Cause-and-Effect Diagram

slide 8

- 1 Define the problem or effect to be analyzed.
- 2 Form the team to perform the analysis (often via brainstorming).
- 3 Draw the effect box and the center line.
- 4 Specify the major potential cause categories and join them as boxes to the center line.
- 5 Identify the possible causes and classify them into the categories; create new categories if necessary.
- 6 Rank-order the causes to identify those most likely to impact the problem.
- 7 Take corrective action.

Chance causes

A certain amount of inherent / natural variability always exists. This natural variability (“**background noise**”) is the cumulative effect of many small, essentially **unavoidable** causes — a “stable system of chance causes.”

✓ In statistical control

A process operating with **only chance causes** of variation present is said to be **in statistical control**.

Assignable causes

Other variability arises mainly from three sources: (1) improperly adjusted or controlled machines, (2) **operator errors**, (3) **defective raw material**. This variability is large compared to the background noise and represents an unacceptable level of performance.

✗ Out of control

A process operating in the presence of **assignable causes** is said to be **out of control**.

Objective of SPC

The major objective is to **quickly detect the occurrence of assignable causes**. The control chart is an **on-line** monitoring technique; it can also estimate process parameters, determine **process capability**, and help improve the process. The eventual goal is the **elimination of variability**.

3 Statistical Basis of the Control Chart

slide 25

A control chart contains three lines:

CL

Center Line

UCL

Upper Control Limit

LCL

Lower Control Limit

- A point that plots **within** the limits → process **in control** → **no action necessary**.
- A point that plots **outside** the limits → evidence the process is **out of control** → investigation and corrective action to find and eliminate the assignable cause(s).

Subtle but important

Even if all points plot inside the limits, if they behave in a **systematic or nonrandom** manner, this could indicate the process is out of control.

Specifying the control limits is one of the critical decisions in designing a control chart — a trade-off between two errors:

Change	Type I error (false alarm)	Type II error (miss a real shift)
Move limits farther from CL	decreases ▼	increases ▲
Move limits closer to CL	increases ▲	decreases ▼

Definitions

Type I (α): a point exceeds the limits and signals out of control *when no assignable cause is present* (false alarm).

Type II (β): a point falls *inside* the limits *when the process is really out of control*.

- **3-Sigma Control Limits:** probability of type I error = **0.0027**.
- **Probability Limits:** the type I error probability is chosen directly (e.g. 0.001 gives 3.09-sigma limits).
- **Warning Limits:** typically selected as **2-sigma** limits.

 Example values (slide 36)

With $CL = 1.5$ and $\sigma_{\bar{x}} \approx 0.0671$:

UCL ($+3\sigma$)	1.7013
UWL ($+2\sigma$, upper warning)	1.6342
CL	1.5000
LWL (-2σ , lower warning)	1.3658
LCL (-3σ)	1.2487

3-sigma limits

$$UCL = CL + 3\sigma \quad \cdot \quad LCL = CL - 3\sigma \quad \cdot \quad (\text{Warning} = CL \pm 2\sigma)$$

- **Larger samples** make it easier to detect **small shifts**.
- Choice between small samples at short intervals vs. larger samples at longer intervals.
- Current industry practice favors **smaller, more frequent samples**.

Average Run Length (ARL)

The ARL is the **average number of points** that must be plotted **before** a point indicates an out-of-control condition. For any Shewhart chart (uncorrelated observations):

Eq 5.2

$$ARL = \frac{1}{p} \quad (5.2)$$

where p = probability that any point **exceeds** the control limits.



In-control ARL (slides 39-40)

For an $\bar{x}$ chart with 3-sigma limits, in control $p = 0.0027$:

$$ARL_0 = \frac{1}{0.0027} = 370$$

So a false alarm occurs about every **370 samples** on average. The run length is **geometric**, with a large standard deviation:

$$\sigma_{RL} = \frac{\sqrt{(1-p)}}{p} \approx 370$$

The 10th and 50th percentiles are **38** and **256** ($\approx 10\%$ of the time the run length ≤ 38 ; 50% of the time ≤ 256).

If samples are taken at fixed intervals h hours apart, performance can be expressed as the **average time to signal**:

Eq 5.3

$$ATS = ARL \times h \quad (5.3)$$

In control, sampling every hour $\rightarrow$ a false alarm about every **370 hours**.

⚠ **The meaning of p depends on the state**

In control: $p = \alpha = 0.0027$ (false-alarm probability) $\rightarrow$ gives **ARL₀**.

Out of control: $p = 1 - \beta$ (probability of **detecting** the shift) $\rightarrow$ gives **ARL₁**.



Detecting a shift — $n = 5$ (slide 42)

The mean shifts to 1.725 microns. From the OC curve, the probability of $\bar{x}$ falling **between** the limits ≈ 0.65 ($= \beta$), so $p = 1 - 0.65 = 0.35$:

$$ARL_1 = \frac{1}{0.35} = 2.86 \rightarrow ATS = 2.86 \times 1 = 2.86 \text{ hr}$$



Faster detection (slides 43-44)

(a) Sample more often: every half hour $\rightarrow$ $ATS = 2.86 \times \frac{1}{2} = \mathbf{1.43 \text{ hr.}}$

(b) Larger sample: $n = 10 \rightarrow P(\text{between}) \approx 0.1 (\beta)$, $p = 0.9 \rightarrow ARL_1 = 1/0.9 = 1.11 \rightarrow ATS = \mathbf{1.11 \text{ hr.}}$

A larger sample detects the shift about **twice as fast**. Two equivalent designs:

	Design 1	Design 2
Sample size	$n = 5$	$n = 10$
Sampling frequency	every half hour	every hour

Rational Subgroups

Subgroups or samples should be selected so that — if assignable causes are present — the chance for differences **between** subgroups is **maximized**, while the chance for differences due to assignable causes **within** a subgroup is **minimized**.

Two general approaches for constructing rational subgroups:

① Units produced at the same time (consecutive units)

Primary purpose: **detect process shifts** (a "snapshot" of the process).

② A random sample representative of all units produced since the last sample

Often used to make decisions about **acceptance of product**; effective at detecting shifts out and back between samples.

⚠ Caution

With approach ②, care is needed: you can make almost any process **appear** to be in statistical control simply by stretching out the interval between observations in the sample.



Formula Cheat Sheet

Quantity	Formula	Note
3-sigma control limits	$UCL = CL + 3\sigma$, $LCL = CL - 3\sigma$	most common
Warning limits	$CL \pm 2\sigma$	2-sigma
Type I error (3σ)	$\alpha = 0.0027$	false alarm
Average run length	$ARL = 1 / p$	Eq 5.2
In-control ARL	$ARL_0 = 1 / 0.0027 = 370$	$p = \alpha$
Run-length std. dev.	$\sigma_{RL} = \sqrt{(1 - p) / p}$	geometric
Average time to signal	$ATS = ARL \times h$	Eq 5.3
Out-of-control ARL	$ARL_1 = 1 / (1 - \beta)$	$p = \text{detection prob.}$



Quick Review

- **Magnificent seven:** Histogram · Check sheet · Pareto · Cause-and-effect · Defect concentration · Scatter · Control chart (most sophisticated).
- **Chance causes** (natural, unavoidable) → in control · **Assignable causes** (machines / operator / material) → out of control.
- Chart lines: **CL · UCL · LCL**; inside = no action, outside = investigate; nonrandom pattern inside the limits is also a signal.
- **Wider limits** ↓ type I & ↑ type II; **narrower** the opposite. ($3\sigma \rightarrow \alpha = 0.0027$.)
- **ARL = $1/p$** · $ARL_0 = 370$ · $ATS = ARL \cdot h$ · $ARL_1 = 1/(1-\beta)$; larger $n \Rightarrow$ faster detection.
- **Rational subgroups:** maximize between-subgroup difference, minimize within-subgroup difference.